Applied Economics

Panel Data Methods

Pello Aspuru

Winter 2025/26

Plan for this Lecture

1. What is Panel Data?
2. Within and Between Variation
3. Panel Data Estimators
    - Dummy Variable Estimator
    - First Differences
    - Within-Estimator

Motivation

Endogeneity due to OVB is a pervasive issue.

One approach to address it: instrumental variable regressions Finding suitable instruments is very hard.

Today we consider an alternative approach to address certain forms of omitted variables: **panel data** allows to control for all *time-invariant* unobservables.

Motivation

Endogeneity due to OVB is a pervasive issue.

One approach to address it: instrumental variable regressions Finding suitable instruments is very hard.

Today we consider an alternative approach to address certain forms of omitted variables: **panel data** allows to control for all *time-invariant* unobservables.

What is Panel Data?

What is Panel Data?

Does seatbelt use cause a decrease in traffic fatalities?

```
df <- read.csv("./seatbelts.csv")
head(df)
```

Panel data: Observe the same cross-sectional unit over multiple time periods

- *cross-sectional unit* can be a person, company, city, state, etc. $(i = 1, \ldots n)$
- *time period* can be a month, a day, a year, etc. $(t = 1, \ldots, T)$

We denote a sample of panel observations as $\{Y_{it}, \boldsymbol{X}_{it}\}$, $i = 1, ..., n$, and $t = 1, ..., T$.

What is Panel Data?

Does seatbelt use cause a decrease in traffic fatalities?

```
df <- read.csv("./seatbelts.csv")
head(df)
```

Panel data: Observe the same cross-sectional unit over multiple time periods

- *cross-sectional unit* can be a person, company, city, state, etc. $(i = 1, \ldots n)$
- *time period* can be a month, a day, a year, etc. $(t = 1, \ldots, T)$

We denote a sample of panel observations as $\{Y_{it}, \boldsymbol{X}_{it}\}$, $i = 1, ..., n$, and $t = 1, ..., T$.

Balanced and Unbalance Panels

- **Balanced** panel: $\{Y_{it}, \boldsymbol{X}_{it}\}$ $i = 1, ..., n; t = 1, ..., T$ [n x T observations in total]
- **Unbalanced** panel: $\{Y_{it}, \boldsymbol{X}_{it}\}$, $i = 1, ..., n; t = \underline{t}_i ..., \overline{t}_i$, where $\underline{t}_i$ and $\overline{t}_i$ may vary across i. $\rightarrow$ we do not observe the same number of periods for each i.

$\rightarrow$ Unbalancedness is prevalent, will abstract from this for the sake of exposition.

Take a look to `df`. Is it balanced or unbalanced?

Balanced and Unbalance Panels

- **Balanced** panel: $\{Y_{it}, \boldsymbol{X}_{it}\}$ $i = 1, ..., n; t = 1, ..., T$ [n x T observations in total]
- **Unbalanced** panel: $\{Y_{it}, \boldsymbol{X}_{it}\}$, $i = 1, ..., n; t = \underline{t}_i ..., \overline{t}_i$, where $\underline{t}_i$ and $\overline{t}_i$ may vary across i. $\rightarrow$ we do not observe the same number of periods for each i.

$\rightarrow$ Unbalancedness is prevalent, will abstract from this for the sake of exposition.

Take a look to `df`. Is it balanced or unbalanced?

Balanced and Unbalance Panels

- **Balanced** panel: $\{Y_{it}, \boldsymbol{X}_{it}\}$ $i = 1, ..., n; t = 1, ..., T$ [n x T observations in total]
- **Unbalanced** panel: $\{Y_{it}, \boldsymbol{X}_{it}\}$, $i = 1, ..., n; t = \underline{t}_i ..., \overline{t}_i$, where $\underline{t}_i$ and $\overline{t}_i$ may vary across i. $\rightarrow$ we do not observe the same number of periods for each i.

$\rightarrow$ Unbalancedness is prevalent, will abstract from this for the sake of exposition.

Take a look to `df`. Is it balanced or unbalanced?

Panel Data - Features and Assumptions

Focus on "large n, small T" panels.

- many cross-sectional units n (which may be randomly sampled)
- observed for small number of time periods T (which are typically ordered)

$\rightarrow$ Small n, larg T panels: time series econometrics, macroeconometrics.

Standard Assumptions:

- cross-sectional units (e.g states) are randomly sampled from a large population $\rightarrow$ i.i.d. across $i = 1, ..., n$.
- At the same time, we will not make the i.i.d assumption within a unit i across time (i.e., for a given unit i), as it is typically implausible in most settings.
  - example: state i car fatalities in year t and year $t + 1$

Panel Data - Features and Assumptions

Focus on "large n, small T" panels.

- many cross-sectional units n (which may be randomly sampled)
- observed for small number of time periods T (which are typically ordered)

$\rightarrow$ Small n, larg T panels: time series econometrics, macroeconometrics.

Standard Assumptions:

- cross-sectional units (e.g states) are randomly sampled from a large population $\rightarrow$ i.i.d. across $i = 1, ..., n$.
- At the same time, we will not make the i.i.d assumption within a unit i across time (i.e., for a given unit i), as it is typically implausible in most settings.
  - example: state i car fatalities in year t and year $t + 1$

Panel Data - Features and Assumptions

Focus on "large n, small T" panels.

- many cross-sectional units n (which may be randomly sampled)
- observed for small number of time periods T (which are typically ordered)

$\rightarrow$ Small n, larg T panels: time series econometrics, macroeconometrics.

Standard Assumptions:

- cross-sectional units (e.g states) are randomly sampled from a large population $\rightarrow$ i.i.d. across $i = 1, ..., n$.
- At the same time, we will not make the i.i.d assumption within a unit i across time (i.e., for a given unit i), as it is typically implausible in most settings.
  - example: state i car fatalities in year t and year $t + 1$

Detour: `dplyr` Package for Data Manipulation

The Pipe Operator `%>%` Reads as "*and then*"

- Passes the left-hand side as the first argument to the function on the right.

Core `dplyr` Verbs:

- `arrange()` – sort rows
- `group_by()` – apply operations within groups
- `summarise()` – compute summary statistics
- `mutate()` – create or transform variables
- Many other: `filter()`, `select()`, ...

Example: find average seatbelt use per state, save as `df_average`:

```
df_average <- df %>% group_by(state)
                 %>% summarise(avg_usage = mean(seatbelt_usage))
```

How many rows and columns will `df_average` have?

Detour: `dplyr` Package for Data Manipulation

The Pipe Operator %>% Reads as "*and then*"

- Passes the left-hand side as the first argument to the function on the right.

Core `dplyr` Verbs:

- `arrange()` – sort rows
- `group_by()` – apply operations within groups
- `summarise()` – compute summary statistics
- `mutate()` – create or transform variables
- Many other: `filter()`, `select()`, ...

Example: find average seatbelt use per state, save as `df_average`:

```
df_average <- df %>% group_by(state)
                 %>% summarise(avg_usage = mean(seatbelt_usage))
```

How many rows and columns will `df_average` have?

Detour: `dplyr` Package for Data Manipulation

The Pipe Operator `%>%` Reads as "*and then*"

- Passes the left-hand side as the first argument to the function on the right.

Core `dplyr` Verbs:

- `arrange()` – sort rows
- `group_by()` – apply operations within groups
- `summarise()` – compute summary statistics
- `mutate()` – create or transform variables
- Many other: `filter()`, `select()`, ...

Example: find average seatbelt use per state, save as `df_average`:

```
df_average <- df %>% group_by(state)
                 %>% summarise(avg_usage = mean(seatbelt_usage))
```

How many rows and columns will `df_average` have?

Detour: dplyr Package for Data Manipulation

The Pipe Operator %>% Reads as "*and then*"

- Passes the left-hand side as the first argument to the function on the right.

Core dplyr Verbs:

- `arrange()` – sort rows
- `group_by()` – apply operations within groups
- `summarise()` – compute summary statistics
- `mutate()` – create or transform variables
- Many other: `filter()`, `select()`, ...

Example: find average seatbelt use per state, save as `df_average`:

```
df_average <- df %>% group_by(state)
                 %>% summarise(avg_usage = mean(seatbelt_usage))
```

How many rows and columns will `df_average` have?

Within and Between Variation

Single Linear Regression - Pooled OLS

Start from a single linear regression model of fatality rates Y_{it} and seatbelt use $X_{1,it}$

$$Y_{it} = \beta_0 + \beta_1 X_{1,it} + u_{it}, \qquad i = 1, ..., n; \qquad t = 1, ..., T.$$

Seatbelt use X_{it} varies between states and within states.

Pooled OLS Estimate the parameters of the model by OLS

```
pooled_slr <- lm(fatality_rate ~ seatbelt_usage, data = df)
coeftest(pooled_slr, vcov. = vcovHC(pooled_slr, "HC0"))
```

- Advantage: Slope parameter β_1 is estimated using between and within variation.
- Disadvantage: Endogeneity is very likely $E[u_{it} X_{it}] \neq 0$

Single Linear Regression - Pooled OLS

Start from a single linear regression model of fatality rates Y_{it} and seatbelt use $X_{1,it}$

$$Y_{it} = \beta_0 + \beta_1 X_{1,it} + u_{it}, \qquad i = 1, ..., n; \qquad t = 1, ..., T.$$

Seatbelt use X_{it} varies between states and within states.

Pooled OLS Estimate the parameters of the model by OLS

```
pooled_slr <- lm(fatality_rate ~ seatbelt_usage, data = df)
coeftest(pooled_slr, vcov. = vcovHC(pooled_slr, "HC0"))
```

- Advantage: Slope parameter β_1 is estimated using between and within variation.
- Disadvantage: Endogeneity is very likely $E[u_{it} X_{it}] \neq 0$

Single Linear Regression - Pooled OLS

Start from a single linear regression model of fatality rates Y_{it} and seatbelt use $X_{1,it}$

$$Y_{it} = \beta_0 + \beta_1 X_{1,it} + u_{it}, \qquad i = 1, ..., n; \qquad t = 1, ..., T.$$

Seatbelt use X_{it} varies between states and within states.

Pooled OLS Estimate the parameters of the model by OLS

```
pooled_slr <- lm(fatality_rate ~ seatbelt_usage, data = df)
coeftest(pooled_slr, vcov. = vcovHC(pooled_slr, "HC0"))
```

- Advantage: Slope parameter β_1 is estimated using between and within variation.
- Disadvantage: Endogeneity is very likely $E[u_{it} X_{it}] \neq 0$

Single Linear Regression Pooled OLS - First Look at Data

Plot the correlation between the main regressor and the outcome variable

```
scatter_1 <-
    ggplot(df, aes(x = seatbelt_usage, y = fatality_rate,
                   color = state)) +
    geom_point(size = 3) +
    geom_smooth(method = "lm", se = FALSE, color = "blue") +
    labs(x = "Seatbelt Usage", y = "Fatality Rate") +
    theme_minimal() +
    theme(legend.position = "bottom")
```

Save plot:

```
ggsave("./AE_week9_panel/scatterplot_1.pdf",
       plot = scatter_1, width = 7, height = 5)
```

Single Linear Regression Pooled OLS - First Look at Data

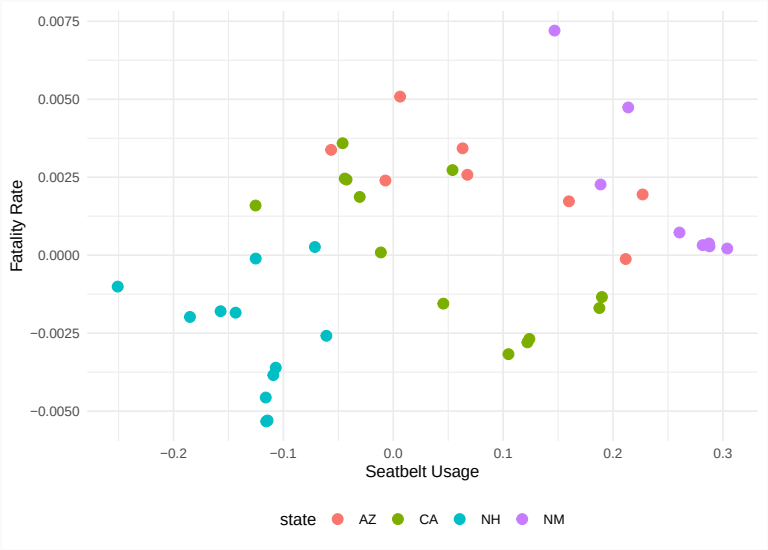
Plot the correlation between the main regressor and the outcome variable

```
scatter_1 <-
    ggplot(df, aes(x = seatbelt_usage, y = fatality_rate,
                   color = state)) +
    geom_point(size = 3) +
    geom_smooth(method = "lm", se = FALSE, color = "blue") +
    labs(x = "Seatbelt Usage", y = "Fatality Rate") +
    theme_minimal() +
    theme(legend.position = "bottom")
```

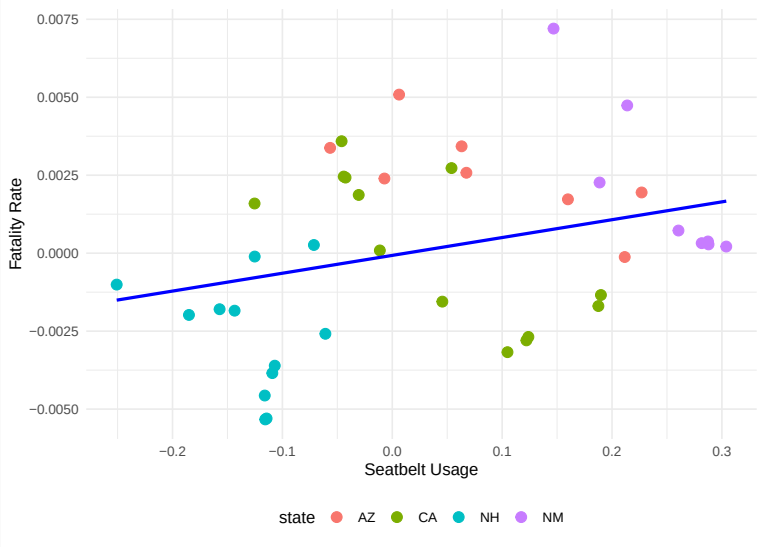
Save plot:

```
ggsave("./AE_week9_panel/scatterplot_1.pdf",
       plot = scatter_1, width = 7, height = 5)
```

Pooled OLS - First Look at Data



Pooled OLS - First Look at Data



Endogeneity

We can tame endogeneity with a pooled multiple linear regression that controls for mean state income $X_{2,it}$ and mean age $X_{3,it}$:

$$Y_{it} = \beta_0 + \beta_1 X_{1,it} + \beta_2 X_{2,it} + \beta_3 X_{3,it} + v_{it}, \qquad i = 1, ..., n; \qquad t = 1, ..., T.$$

```
pooled_mlr <- lm(fatality_rate ~ seatbelt_usage + income + age,
                 data = df)
coeftest(pooled_mlr, vcov. = vcovHC(pooled_mlr, "HC0"))
```

- Advantage: Slope parameters β are estimated using between and within variation.
- Disadvantage: Endogeneity is very likely $E[v_{it} X_{it}] \neq 0$

$\rightarrow$ Previous advantages and disadvantages remain.

Panel Data Method intuition:
sacrifice the between variation in exchange of less endogeneity.

Endogeneity

We can tame endogeneity with a pooled multiple linear regression that controls for mean state income $X_{2,it}$ and mean age $X_{3,it}$:

$$Y_{it} = \beta_0 + \beta_1 X_{1,it} + \beta_2 X_{2,it} + \beta_3 X_{3,it} + v_{it}, \qquad i = 1, ..., n; \qquad t = 1, ..., T.$$

```
pooled_mlr <- lm(fatality_rate ~ seatbelt_usage + income + age,
                 data = df)
coeftest(pooled_mlr, vcov. = vcovHC(pooled_mlr, "HC0"))
```

- Advantage: Slope parameters β are estimated using between and within variation.
- Disadvantage: Endogeneity is very likely $E[v_{it} X_{it}] \neq 0$

$\rightarrow$ Previous advantages and disadvantages remain.

Panel Data Method intuition:
sacrifice the between variation in exchange of less endogeneity.

Endogeneity

We can tame endogeneity with a pooled multiple linear regression that controls for mean state income $X_{2,it}$ and mean age $X_{3,it}$:

$$Y_{it} = \beta_0 + \beta_1 X_{1,it} + \beta_2 X_{2,it} + \beta_3 X_{3,it} + v_{it}, \qquad i = 1, ..., n; \qquad t = 1, ..., T.$$

```
pooled_mlr <- lm(fatality_rate ~ seatbelt_usage + income + age,
                 data = df)
coeftest(pooled_mlr, vcov. = vcovHC(pooled_mlr, "HC0"))
```

- Advantage: Slope parameters β are estimated using between and within variation.
- Disadvantage: Endogeneity is very likely $E[v_{it} X_{it}] \neq 0$

$\rightarrow$ Previous advantages and disadvantages remain.

Panel Data Method intuition:

sacrifice the between variation in exchange of less endogeneity.

Within- vs. between-variation

Pooled OLS treats the whole data as one large (pooled) i.i.d. sample.

- often not justified as observations are not independent within units i across t.
- moreover, it does not exploit the specific advantages of panel data!

In a panel data set, we can decompose the total variation in our variables into

1. within variation: changes within each cross-sectional group i over time.
   $\rightarrow$ Used to estimate slope parameter β_1.
2. between variation: changes between cross-sectional groups
   $\rightarrow$ Used to reduce endogeneity.

Example: seat belt usage rate varies *between* states, and *within* each state across years. minimum drinking age only varies between states. It is time-invariant.

Within- vs. between-variation

Pooled OLS treats the whole data as one large (pooled) i.i.d. sample.

- often not justified as observations are not independent within units i across t.
- moreover, it does not exploit the specific advantages of panel data!

In a panel data set, we can decompose the total variation in our variables into

1. within variation: changes within each cross-sectional group i over time.
   $\rightarrow$ Used to estimate slope parameter β_1.
2. between variation: changes between cross-sectional groups
   $\rightarrow$ Used to reduce endogeneity.

Example: seat belt usage rate varies *between* states, and *within* each state across years. minimum drinking age only varies between states. It is time-invariant.

The Panel Model

Key assumption: **decompose** unobserved error term u_{it} into two:

- time-invariant component c_i: only varies between units i.
- idiosyncratic error term ϵ_{it} that varies between *and* within unit i over time.

The decomposition allows us to rewrite the single linear regression model as:

$$Y_{it} = \beta_0 + \beta_1 X_{1,it} + \underbrace{c_i + \epsilon_{it}}_{u_{it}}, i = 1, ..., n; t = 1, ..., T.$$

Assuming that c_i is potentially correlated with included regressors $\boldsymbol{X}_{it}$:
$Cov(\boldsymbol{X}_{it}, c_i) \neq 0 \Rightarrow E[\epsilon_{it} + c_i | \boldsymbol{X}_{it}] \neq 0 \rightarrow$ endogeneity, bias in pooled OLS

Pooled OLS exogeneity $E[u_{it}|\boldsymbol{X_{it}}] = 0 \rightarrow$ Panel exogeneity $E[\epsilon_{it}|\boldsymbol{X_{it}}, c_i] = 0$

We control for all **time-invariant** endogeneity

The Panel Model

Key assumption: **decompose** unobserved error term u_{it} into two:

- time-invariant component c_i: only varies between units i.
- idiosyncratic error term ϵ_{it} that varies between *and* within unit i over time.

The decomposition allows us to rewrite the single linear regression model as:

$$Y_{it} = \beta_0 + \beta_1 X_{1,it} + \underbrace{c_i + \epsilon_{it}}_{u_{it}}, i = 1, ..., n; t = 1, ..., T.$$

Assuming that c_i is potentially correlated with included regressors $\boldsymbol{X}_{it}$:
$Cov(\boldsymbol{X}_{it}, c_i) \neq 0 \Rightarrow E[\epsilon_{it} + c_i | \boldsymbol{X}_{it}] \neq 0 \rightarrow$ endogeneity, bias in pooled OLS

Pooled OLS exogeneity $E[u_{it}|\boldsymbol{X_{it}}] = 0 \rightarrow$ Panel exogeneity $E[\epsilon_{it}|\boldsymbol{X_{it}}, c_i] = 0$

We control for all **time-invariant** endogeneity

The Panel Model

Key assumption: **decompose** unobserved error term u_{it} into two:

- time-invariant component c_i: only varies between units i.
- idiosyncratic error term ϵ_{it} that varies between *and* within unit i over time.

The decomposition allows us to rewrite the single linear regression model as:

$$Y_{it} = \beta_0 + \beta_1 X_{1,it} + \underbrace{c_i + \epsilon_{it}}_{u_{it}}, i = 1, ..., n; t = 1, ..., T.$$

Assuming that c_i is potentially correlated with included regressors $\boldsymbol{X}_{it}$:
$Cov(\boldsymbol{X}_{it}, c_i) \neq 0 \Rightarrow E[\epsilon_{it} + c_i | \boldsymbol{X}_{it}] \neq 0 \rightarrow$ endogeneity, bias in pooled OLS

Pooled OLS exogeneity $E[u_{it}|\boldsymbol{X}_{it}] = 0 \rightarrow$ Panel exogeneity $E[\epsilon_{it}|\boldsymbol{X}_{it}, c_i] = 0$

We control for all **time-invariant** endogeneity

The Panel Model

Key assumption: **decompose** unobserved error term u_{it} into two:

- time-invariant component c_i: only varies between units i.
- idiosyncratic error term ϵ_{it} that varies between *and* within unit i over time.

The decomposition allows us to rewrite the single linear regression model as:

$$Y_{it} = \beta_0 + \beta_1 X_{1,it} + \underbrace{c_i + \epsilon_{it}}_{u_{it}}, i = 1, ..., n; t = 1, ..., T.$$

Assuming that c_i is potentially correlated with included regressors $\boldsymbol{X}_{it}$:
$Cov(\boldsymbol{X}_{it}, c_i) \neq 0 \Rightarrow E[\epsilon_{it} + c_i | \boldsymbol{X}_{it}] \neq 0 \rightarrow$ endogeneity, bias in pooled OLS

Pooled OLS exogeneity $E[u_{it}|\boldsymbol{X_{it}}] = 0 \rightarrow$ Panel exogeneity $E[\epsilon_{it}|\boldsymbol{X_{it}}, c_i] = 0$

We control for all **time-invariant** endogeneity

The Panel Model

Key assumption: **decompose** unobserved error term u_{it} into two:

- time-invariant component c_i: only varies between units i.
- idiosyncratic error term ϵ_{it} that varies between *and* within unit i over time.

The decomposition allows us to rewrite the single linear regression model as:

$$Y_{it} = \beta_0 + \beta_1 X_{1,it} + \underbrace{c_i + \epsilon_{it}}_{u_{it}}, i = 1, ..., n; t = 1, ..., T.$$

Assuming that c_i is potentially correlated with included regressors $\boldsymbol{X}_{it}$:
$Cov(\boldsymbol{X}_{it}, c_i) \neq 0 \Rightarrow E[\epsilon_{it} + c_i | \boldsymbol{X}_{it}] \neq 0 \rightarrow$ endogeneity, bias in pooled OLS

Pooled OLS exogeneity $E[u_{it}|\boldsymbol{X_{it}}] = 0 \rightarrow$ Panel exogeneity $E[\epsilon_{it}|\boldsymbol{X_{it}}, c_i] = 0$

We control for all **time-invariant** endogeneity

The Panel Model

Key assumption: **decompose** unobserved error term u_{it} into two:

- time-invariant component c_i: only varies between units i.
- idiosyncratic error term ϵ_{it} that varies between *and* within unit i over time.

The decomposition allows us to rewrite the single linear regression model as:

$$Y_{it} = \beta_0 + \beta_1 X_{1,it} + \underbrace{c_i + \epsilon_{it}}_{u_{it}}, i = 1, ..., n; t = 1, ..., T.$$

Assuming that c_i is potentially correlated with included regressors $\boldsymbol{X}_{it}$:
$Cov(\boldsymbol{X}_{it}, c_i) \neq 0 \Rightarrow E[\epsilon_{it} + c_i | \boldsymbol{X}_{it}] \neq 0 \rightarrow$ endogeneity, bias in pooled OLS

Pooled OLS exogeneity $E[u_{it}|\boldsymbol{X_{it}}] = 0 \rightarrow$ Panel exogeneity $E[\epsilon_{it}|\boldsymbol{X_{it}}, c_i] = 0$

We control for all **time-invariant** endogeneity

Panel Data Estimators

How do we estimate conrol for the time-invariant endogeneity c_i ?

Three different ways to estimate the **panel model**:

1. The Dummy Variable Regression
2. First Differences Estimator
3. Within Transformation

How do we estimate conrol for the time-invariant endogeneity c_i ?
Three different ways to estimate the **panel model**:

1. The Dummy Variable Regression
2. First Differences Estimator
3. Within Transformation

Option 1: The Dummy Variable Regression

Explicitly add c_i to our regression by adding a dummy for each cross-sectional unit i.[1]

$$Y_{it} = \cancel{\beta_0}^{0} + \beta_1 X_{1,it} + \underbrace{\alpha_1 \mathbb{1}\{state = 1\}}_{c_1} + ... + \underbrace{\alpha_n \mathbb{1}\{state = n\}}_{c_n} + \epsilon_{it}$$

- $E[c_i|X_{1,it}] = 0$ **no longer** required for exogeneity. Examples?
- $E[\epsilon_{it}|X_{1,it}, c_i] = 0$ still required for exogeneity. Examples?

```
panel_dummy <- lm(fatality_rate ~ 0 + factor(state) +
                    seatbelt_usage, data = df)
```

→ Which state has the largest c_i ?

[1] Note: we either have to omit one dummy, or the constant β_0

Option 1: The Dummy Variable Regression

Explicitly add c_i to our regression by adding a dummy for each cross-sectional unit i.[1]

$$Y_{it} = \cancel{\beta_0}^{0} + \beta_1 X_{1,it} + \underbrace{\alpha_1 \mathbb{1}\{state = 1\}}_{c_1} + ... + \underbrace{\alpha_n \mathbb{1}\{state = n\}}_{c_n} + \epsilon_{it}$$

- $E[c_i|X_{1,it}] = 0$ **no longer** required for exogeneity. Examples?
- $E[\epsilon_{it}|X_{1,it}, c_i] = 0$ still required for exogeneity. Examples?

```
panel_dummy <- lm(fatality_rate ~ 0 + factor(state) +
                    seatbelt_usage, data = df)
```

$\rightarrow$ Which state has the largest c_i ?

[1] Note: we either have to omit one dummy, or the constant β_0

Option 1: The Dummy Variable Regression

Explicitly add c_i to our regression by adding a dummy for each cross-sectional unit i.[1]

$$Y_{it} = \cancel{\beta_0}^{0} + \beta_1 X_{1,it} + \underbrace{\alpha_1 \mathbb{1}\{state = 1\}}_{c_1} + ... + \underbrace{\alpha_n \mathbb{1}\{state = n\}}_{c_n} + \epsilon_{it}$$

- $E[c_i|X_{1,it}] = 0$ **no longer** required for exogeneity. Examples?
- $E[\epsilon_{it}|X_{1,it}, c_i] = 0$ still required for exogeneity. Examples?

```
panel_dummy <- lm(fatality_rate ~ 0 + factor(state) +
                    seatbelt_usage, data = df)
```

→ Which state has the largest c_i ?

[1] Note: we either have to omit one dummy, or the constant β_0

Option 1: The Dummy Variable Regression

Explicitly add c_i to our regression by adding a dummy for each cross-sectional unit i.[1]

$$Y_{it} = \cancelto{0}{\beta_0} + \beta_1 X_{1,it} + \underbrace{\alpha_1 \mathbb{1}\{state = 1\}}_{c_1} + ... + \underbrace{\alpha_n \mathbb{1}\{state = n\}}_{c_n} + \epsilon_{it}$$

- $E[c_i|X_{1,it}] = 0$ **no longer** required for exogeneity. Examples?
- $E[\epsilon_{it}|X_{1,it}, c_i] = 0$ still required for exogeneity. Examples?

```
panel_dummy <- lm(fatality_rate ~ 0 + factor(state) +
                    seatbelt_usage, data = df)
```

→ Which state has the largest c_i ?

[1] Note: we either have to omit one dummy, or the constant β_0

Option 1: The Dummy Variable Regression

Explicitly add c_i to our regression by adding a dummy for each cross-sectional unit i.[1]

$$Y_{it} = \cancelto{0}{\beta_0} + \beta_1 X_{1,it} + \underbrace{\alpha_1 \mathbb{1}\{state = 1\}}_{c_1} + ... + \underbrace{\alpha_n \mathbb{1}\{state = n\}}_{c_n} + \epsilon_{it}$$

- $E[c_i|X_{1,it}] = 0$ **no longer** required for exogeneity. Examples?
- $E[\epsilon_{it}|X_{1,it}, c_i] = 0$ still required for exogeneity. Examples?

```
panel_dummy <- lm(fatality_rate ~ 0 + factor(state) +
                    seatbelt_usage, data = df)
```

$\rightarrow$ Which state has the largest c_i ?

[1] Note: we either have to omit one dummy, or the constant β_0

Discussion

We can estimate the model with the cross-sectional-unit-dummies by OLS to obtain $\hat{\beta}$.
$\rightarrow$ OLS is unbiased if $E[\epsilon_{it}|\boldsymbol{X_{it}}, c_i] = 0$ (Assumption 4 for this model).

Because all time-invariant unobservables are eliminated from ϵ_{it}, this is a much less strong assumption than $E[u_{it}|\boldsymbol{X_{it}}] = 0$.

Example: weather conditions much more likely to satisfy conditional mean independence than population's attitude towards seat belts.

When thinking about omitted variables that might be correlated with included regressors, we can ignore all time-invariant unobservables. Every conceivable time-invariant variable is accounted for in our analyses!

Interpretation

Pooled OLS:
Comparing seat belt usage rate and fatality rates *across states and years*, we find that an increase in the seatbelt usage rate by 0.1 (10 percentage points) is associated with an *increase* in the number of fatalities per million miles driven by .00057. (=2.9% of the sample mean).

Dummy variable estimation:
Comparing seat belt usage rate and fatality rates *within states over time*, we find that an increase in the seat belt usage rate by .1 is associated with a *decrease* in the number of fatalities per million miles driven by .0017. (=8.6% of the sample mean).

Interpretation

Pooled OLS:
Comparing seat belt usage rate and fatality rates *across states and years*, we find that an increase in the seatbelt usage rate by 0.1 (10 percentage points) is associated with an *increase* in the number of fatalities per million miles driven by .00057. (=2.9% of the sample mean).

Dummy variable estimation:
Comparing seat belt usage rate and fatality rates *within states over time*, we find that an increase in the seat belt usage rate by .1 is associated with a *decrease* in the number of fatalities per million miles driven by .0017. (=8.6% of the sample mean).

Illustrating the Panel Model

Use dplyr to generate an auxiliar dataset pred_df fitting same β_1 within each state:

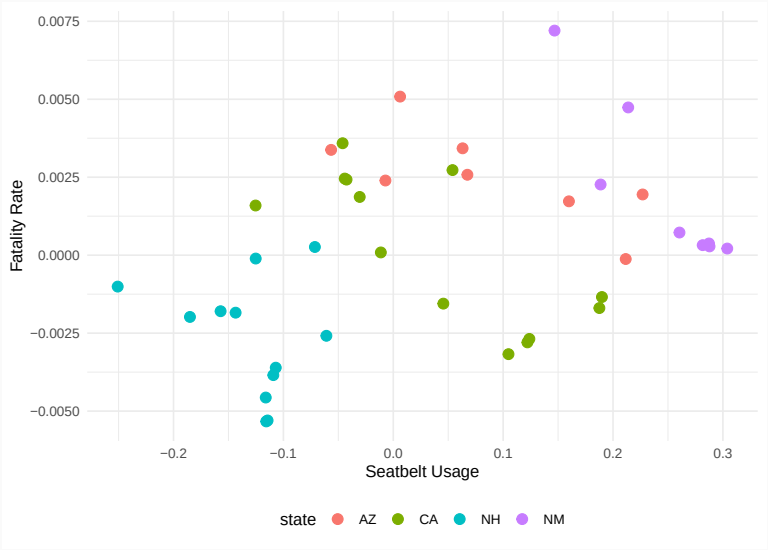
```
pred_df <- df %>%
group_by(state) %>%
summarize(
    seatbelt_usage = seq(min(seatbelt_usage),
                         max(seatbelt_usage),
                         length.out = 100),
    .groups = "drop" # remove previous grouping.
)
pred_df$pred <- predict(panel_dummy, newdata = pred_df)
```

Illustrating the Panel Model

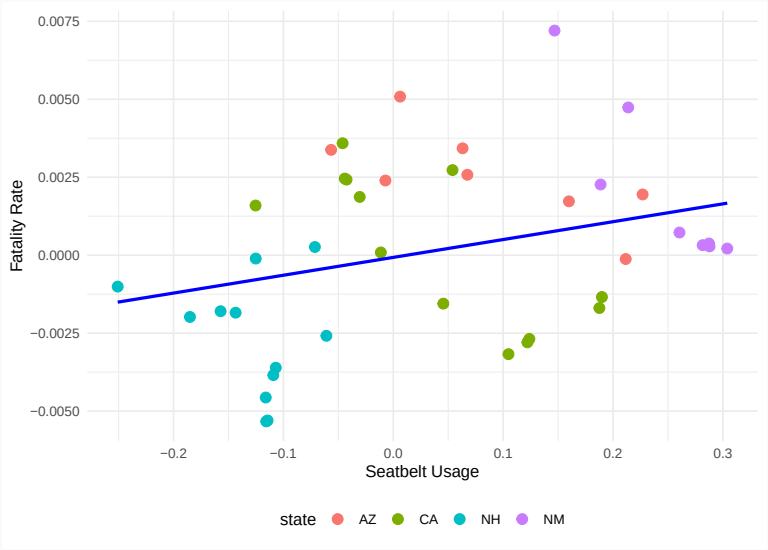
Use ggplot2 to generate scatterplot scatter_2:

```
scatter_2 <-
ggplot(df, aes(x = seatbelt_usage, y = fatality_rate,
       color = state)) +
geom_point(size = 3) +
geom_line(data = pred_df, aes(y = pred), linewidth = 1) +
labs(x = "Seatbelt Usage (%)", y = "Fatality Rate") +
theme_minimal() +
theme(legend.position = "bottom")
```

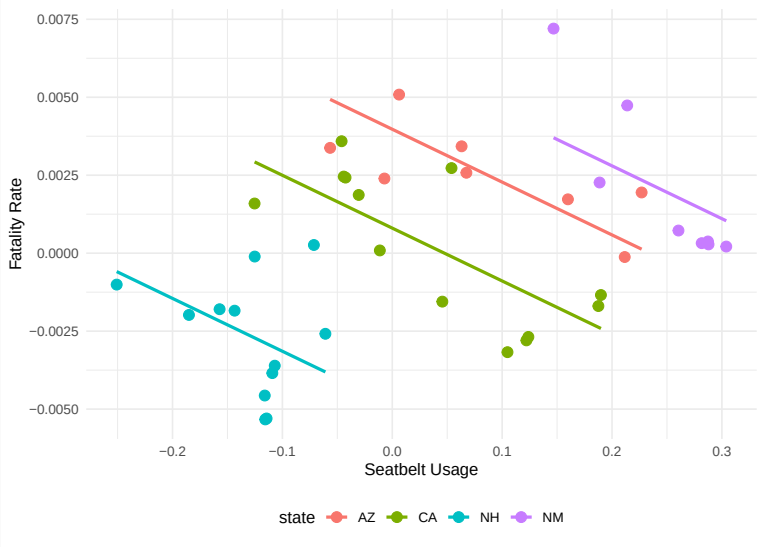
Pooled OLS - Scatterplot



Pooled OLS - Scatterplot



Panel Model - Scatterplot



Option 2: First-Differences Estimator

If $T = 2$, an alternative to the dummy variable approach is first-differencing:

Consider a generic MLR model for two time periods:

$$Y_{i,t=1} = \boldsymbol{X}'_{i,t=1}\boldsymbol{\beta} + c_i + \epsilon_{i,t=1} \text{ and}$$
$$Y_{i,t=2} = \boldsymbol{X}'_{i,t=2}\boldsymbol{\beta} + c_i + \epsilon_{i,t=2}$$

Differencing yields:

$$Y_{i,t=2} - Y_{i,t=1} = (\boldsymbol{X}'_{i,t=2} - \boldsymbol{X}'_{i,t=1})\beta + (c_i - c_i) + (\epsilon_{i,t=2} - \epsilon_{i,t=1})$$
$$\text{or: } \Delta Y_i = \Delta \boldsymbol{X}_i \beta + \Delta \epsilon_i$$

$\rightarrow$ The unobserved heterogeneity c_i drops out! (and also β_0)

This is our "estimating equation".
We can estimate β by OLS by regressing ΔY_i on $\Delta \boldsymbol{X}_i$.
The estimator is unbiased if $E[\Delta \epsilon_i | \Delta \boldsymbol{X}_i] = 0$ (MLR.4 for this model).

Option 2: First-Differences Estimator

If $T = 2$, an alternative to the dummy variable approach is first-differencing:

Consider a generic MLR model for two time periods:

$$Y_{i,t=1} = \boldsymbol{X}'_{i,t=1}\boldsymbol{\beta} + c_i + \epsilon_{i,t=1} \text{ and}$$
$$Y_{i,t=2} = \boldsymbol{X}'_{i,t=2}\boldsymbol{\beta} + c_i + \epsilon_{i,t=2}$$

Differencing yields:

$$Y_{i,t=2} - Y_{i,t=1} = (\boldsymbol{X}'_{i,t=2} - \boldsymbol{X}'_{i,t=1})\boldsymbol{\beta} + (c_i - c_i) + (\epsilon_{i,t=2} - \epsilon_{i,t=1})$$
$$\text{or: } \Delta Y_i = \Delta \boldsymbol{X}_i \boldsymbol{\beta} + \Delta \epsilon_i$$

$\rightarrow$ The unobserved heterogeneity c_i drops out! (and also β_0)

This is our "estimating equation".
We can estimate $\boldsymbol{\beta}$ by OLS by regressing ΔY_i on $\Delta \boldsymbol{X}_i$.
The estimator is unbiased if $E[\Delta \epsilon_i | \Delta \boldsymbol{X}_i] = 0$ (MLR.4 for this model).

Option 2: First-Differences Estimator

If $T = 2$, an alternative to the dummy variable approach is first-differencing:

Consider a generic MLR model for two time periods:

$$Y_{i,t=1} = \boldsymbol{X}'_{i,t=1}\boldsymbol{\beta} + c_i + \epsilon_{i,t=1} \text{ and}$$
$$Y_{i,t=2} = \boldsymbol{X}'_{i,t=2}\boldsymbol{\beta} + c_i + \epsilon_{i,t=2}$$

Differencing yields:

$$Y_{i,t=2} - Y_{i,t=1} = (\boldsymbol{X}'_{i,t=2} - \boldsymbol{X}'_{i,t=1})\boldsymbol{\beta} + (c_i - c_i) + (\epsilon_{i,t=2} - \epsilon_{i,t=1})$$
$$\text{or: } \Delta Y_i = \Delta \boldsymbol{X}_i \boldsymbol{\beta} + \Delta \epsilon_i$$

$\rightarrow$ The unobserved heterogeneity c_i drops out! (and also β_0)

This is our "estimating equation".
We can estimate $\boldsymbol{\beta}$ by OLS by regressing ΔY_i on $\Delta \boldsymbol{X}_i$.
The estimator is unbiased if $E[\Delta \epsilon_i | \Delta \boldsymbol{X}_i] = 0$ (MLR.4 for this model).

Option 2: First-Differences Estimator

If $T = 2$, an alternative to the dummy variable approach is first-differencing:

Consider a generic MLR model for two time periods:

$$Y_{i,t=1} = \boldsymbol{X}'_{i,t=1}\boldsymbol{\beta} + c_i + \epsilon_{i,t=1} \text{ and}$$
$$Y_{i,t=2} = \boldsymbol{X}'_{i,t=2}\boldsymbol{\beta} + c_i + \epsilon_{i,t=2}$$

Differencing yields:

$$Y_{i,t=2} - Y_{i,t=1} = (\boldsymbol{X}'_{i,t=2} - \boldsymbol{X}'_{i,t=1})\boldsymbol{\beta} + (c_i - c_i) + (\epsilon_{i,t=2} - \epsilon_{i,t=1})$$
$$\text{or: } \Delta Y_i = \Delta \boldsymbol{X}_i \boldsymbol{\beta} + \Delta \epsilon_i$$

$\rightarrow$ The unobserved heterogeneity c_i drops out! (and also β_0)

This is our "estimating equation".
We can estimate $\boldsymbol{\beta}$ by OLS by regressing ΔY_i on $\Delta \boldsymbol{X}_i$.
The estimator is unbiased if $E[\Delta \epsilon_i | \Delta \boldsymbol{X}_i] = 0$ (MLR.4 for this model).

Option 3: Within Transformation

1. For all variables in your model, compute the **time-mean** for each unit i:

$$\overline{Y}_i = \frac{1}{T}\sum_{t=1}^{T} Y_{it}, \qquad \overline{X}_{1i} = \frac{1}{T}\sum_{t=1}^{T} X_{1it} \quad \cdots \quad \overline{c_i} = c_i$$

2. Deduct the time means from the actual values. Obtain *demeaned* variables:

$$(Y_{it} - \overline{Y}_i), \quad (X_{1it} - \overline{X}_{1i}), \quad \ldots, \quad (X_{kit} - \overline{X}_{ki}), \quad (c_i - \overline{c_i})$$

Use `dplyr` to generate a demeaned variable dataset `df_demeaned`:

```
df_demeaned <- df %>%
group_by(state) %>%
mutate(fatality_rate_dm   = fatality_rate - mean(fatality_rate),
       seatbelt_usage_dm  = seatbelt_usage - mean(seatbelt_usage)) %>%
ungroup()
```

Option 3: Within Transformation

1. For all variables in your model, compute the **time-mean** for each unit i:

$$\overline{Y}_i = \frac{1}{T}\sum_{t=1}^{T} Y_{it}, \qquad \overline{X}_{1i} = \frac{1}{T}\sum_{t=1}^{T} X_{1it} \quad \cdots \quad \overline{c_i} = c_i$$

2. Deduct the time means from the actual values. Obtain *demeaned* variables:

$$(Y_{it} - \overline{Y}_i), \quad (X_{1it} - \overline{X}_{1i}), \quad \ldots, \quad (X_{kit} - \overline{X}_{ki}), \quad (c_i - \overline{c_i})$$

Use `dplyr` to generate a demeaned variable dataset `df_demeaned`:

```
df_demeaned <- df %>%
group_by(state) %>%
mutate(fatality_rate_dm    = fatality_rate - mean(fatality_rate),
       seatbelt_usage_dm  = seatbelt_usage - mean(seatbelt_usage)) %>%
ungroup()
```

Option 3: Within Transformation

1. For all variables in your model, compute the **time-mean** for each unit i:

$$\overline{Y}_i = \frac{1}{T}\sum_{t=1}^{T} Y_{it}, \qquad \overline{X}_{1i} = \frac{1}{T}\sum_{t=1}^{T} X_{1it} \quad \cdots \quad \overline{c_i} = c_i$$

2. Deduct the time means from the actual values. Obtain *demeaned* variables:

$$(Y_{it} - \overline{Y}_i), \quad (X_{1it} - \overline{X}_{1i}), \quad \ldots, \quad (X_{kit} - \overline{X}_{ki}), \quad (c_i - \overline{c_i})$$

Use `dplyr` to generate a demeaned variable dataset `df_demeaned`:

```
df_demeaned <- df %>%
group_by(state) %>%
mutate(fatality_rate_dm    = fatality_rate - mean(fatality_rate),
       seatbelt_usage_dm  = seatbelt_usage - mean(seatbelt_usage)) %>%
ungroup()
```

Option 3: Within Transformation

Demeaned regression Model:

$$\begin{aligned} Y_{it} - \overline{Y}_i &= (\beta_0 - \beta_0) + \beta_1(X_{1it} - \overline{X}_{1i}) + \cdots + \beta_k(X_{kit} - \overline{X}_{ki}) + (c_i - \overline{c_i}) + (\epsilon_{it} - \overline{\epsilon}_i) \\ &= \beta_1(X_{1it} - \overline{X}_{1i}) + \cdots + \beta_k(X_{kit} - \overline{X}_{ki}) + (\epsilon_{it} - \overline{\epsilon}_i) \end{aligned}$$

- c_i drops out because there is no *within-variation* in c_i!
- Again, we can estimate this "demeaned equation" by OLS.

```
panel_within_manual <- lm(fatality_rate_dm ~ 0 + seatbelt_usage_dm,
                          data = df_demeaned)
coeftest(panel_within_manual,
         vcov. = vcovHC(panel_within_manual, "HC0"))
```

Option 3: Within Transformation

Demeaned regression Model:

$$\begin{aligned} Y_{it} - \overline{Y}_i &= (\beta_0 - \beta_0) + \beta_1(X_{1it} - \overline{X}_{1i}) + \cdots + \beta_k(X_{kit} - \overline{X}_{ki}) + (c_i - \overline{c_i}) + (\epsilon_{it} - \overline{\epsilon}_i) \\ &= \beta_1(X_{1it} - \overline{X}_{1i}) + \cdots + \beta_k(X_{kit} - \overline{X}_{ki}) + (\epsilon_{it} - \overline{\epsilon}_i) \end{aligned}$$

- c_i drops out because there is no *within-variation* in c_i!
- Again, we can estimate this "demeaned equation" by OLS.

```
panel_within_manual <- lm(fatality_rate_dm ~ 0 + seatbelt_usage_dm,
                          data = df_demeaned)
coeftest(panel_within_manual,
         vcov. = vcovHC(panel_within_manual, "HC0"))
```

Within Estimator with `fixest`

We can directly implement the within-estimator in R using the command `feols` from the package `fixest`.

```
panel_within_fixest <- feols(fatality_rate ~ seatbelt_usage | state,
                             data = df)
coeftest(panel_within_fixest,
         vcov. = vcovHC(panel_within_fixest, "HC0"))
```

Within Estimator with `fixest`

We can directly implement the within-estimator in R using the command `feols` from the package `fixest`.

```
panel_within_fixest <- feols(fatality_rate ~ seatbelt_usage | state,
                             data = df)
coeftest(panel_within_fixest,
         vcov. = vcovHC(panel_within_fixest, "HC0"))
```

Limitations of Panel Data Estimators

Recall that, unlike pooled OLS, Panel Data models exploit only *within*-variation for each unit i, i.e. **deviations from the unit-specific mean**.

- We cannot include time-invariant regressors when using these estimators.
  - They would drop out, because they are absorbed by the individual fixed effect.
  - In our example, we could not include a maximum speed control, right?
- For setups with little within group time-variation, we may not have enough variation to estimate slope parameters.

Limitations of Panel Data Estimators

Recall that, unlike pooled OLS, Panel Data models exploit only *within*-variation for each unit i, i.e. **deviations from the unit-specific mean**.

- We cannot include time-invariant regressors when using these estimators.
  - They would drop out, because they are absorbed by the individual fixed effect.
  - In our example, we could not include a maximum speed control, right?
- For setups with little within group time-variation, we may not have enough variation to estimate slope parameters.

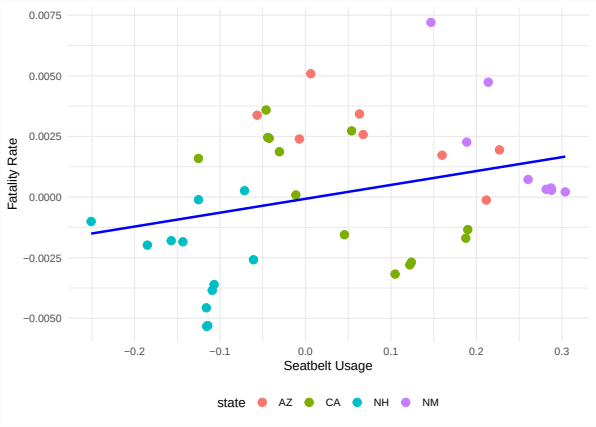
Limitations of Panel Data Estimators

Recall that, unlike pooled OLS, Panel Data models exploit only *within*-variation for each unit i, i.e. **deviations from the unit-specific mean**.

- We cannot include time-invariant regressors when using these estimators.
  - They would drop out, because they are absorbed by the individual fixed effect.
  - In our example, we could not include a maximum speed control, right?
- For setups with little within group time-variation, we may not have enough variation to estimate slope parameters.

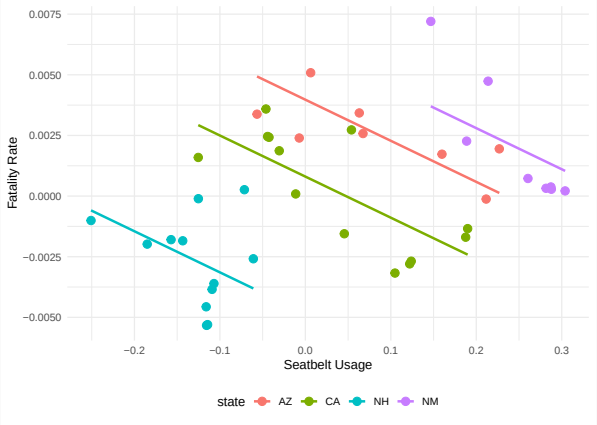
One More Thing...

$$Y_{it} = \beta_0 + \beta_1 X_{1,it} + u_{it}, \quad i = 1, \ldots, n; \quad t = 1, \ldots, T.$$



One More Thing...

$$Y_{it} - \overline{Y}_i = \beta_1(X_{1it} - \overline{X}_{1i}) + (\epsilon_{it} - \overline{\epsilon}_i)$$



One More Thing...

$$Y_{it} - \overline{Y}_i = \beta_1(X_{1it} - \overline{X}_{1i}) + (\epsilon_{it} - \overline{\epsilon}_i)$$

